

PhD-CS Annual Assessment Report

- **Program Name:** Doctor of Philosophy (PhD) in Computer Science
- **Department Name:** Computer Science
- **College Name:** College of Computing
- **Faculty Assessment Coordinator:** CS Graduate Committee
- **Assessment Cycle:** Academic Year 2024–25

1. Program Learning Goals for 2024–25

- **P1.** Have advanced core knowledge in theory.
- **P2.** Have advanced core knowledge in systems.
- **P3.** Have advanced core knowledge in languages.
- **P4.** Have the ability to conduct research (review literature, formulate scientific research questions, collect theoretical or empirical support, present research findings).

2. Assessment Methodology

Learning Goal	Sample Size (Enrollment)	Semester Collected	Rubric Used	Artifact Source (Course/Instructor/Assignment/Other)	Month/Year of Evaluation	Evaluator(s)
P1: Theory	3	Spring 2025	Theory Rubric (Appendix A)	CS535 – Design and Analysis of Algorithms (Instructor: Peng-Jun Wan)	May 2025	Sanjiv Kapoor
P2: Systems	8	Spring 2025	Systems Rubric (Appendix B)	CS550 – Advanced Operating Systems (Instructor: Xian-He Sun)	May 2025	Oliver Michel
P3: Languages	6	Spring 2025	Programming Languages Rubric (Appendix C)	CS536 – Science of Programming (Instructor: Xiaolang Wang)	May 2025	Yutong Wang
P4: Research	7	Fall 2024 + Spring 2025	Research/Qualifier Rubric (Appendix D)	Oral Qualifier Exam (committee evaluation)	May 2025	Bogdan Korel

3. Assessment Results

Learning Goal P1: Theory

- **Average performance:** 85% overall achievement.
- Strongest topics: Shortest paths, backtracking, dynamic programming, approximation algorithms (100%).
- Weaker topics: Graph traversal, MST/arborescence, bipartite matching, max-flow (67%).
- **Grades:** 2 students earned A, 1 earned B.

Discussion: Students demonstrated solid ability to translate real-world problems into optimization problems and provide algorithmic solutions. A small sample size (3) limits generalization, but performance indicates readiness for PhD-level research.

Learning Goal P2: Systems

- **Overall achievement:** 87–95% across assignments, projects, and exams.
- Strongest performance: Centralized/distributed file systems (92–95%).
- Weaker area: Final exam (87%).
- All PhD students achieved A-level performance, meeting the higher standard required.

Discussion: Students demonstrated mastery in systems design and implementation. Results confirm that the PhD cohort meets expectations for advanced systems knowledge.

Learning Goal P3: Languages

- **Overall performance:** Mixed, with a wide distribution.
- Strongest areas: Predicate logic verification, proofs, error avoidance (83%).
- Weaker areas: Hoare triples, loop correctness, parallel programming (50–67%).
- Lowest achievement: Substitutions (33%), program convergence (33%).

Discussion: Many students struggled due to insufficient preparation in discrete mathematics and logic. Suggested remedies include updating prerequisites (CS330/CS440 material integrated into CS401 or requiring CS430). Enrollment was small (6), so percentages are sensitive to individual performance.

Learning Goal P4: Research

- **Overall performance:** 85% of students passed (6 of 7).
- Strongest areas: Literature review and research question formulation (both 85%).
- Weaker areas: Collecting empirical/theoretical support and presenting findings (85% but noted as relative weaknesses).
- Failures: 1 student did not pass.

Discussion: Students are generally capable of research-level work, but presentation and communication of findings remain weaker. Advisors are encouraged to incorporate **practice presentations with feedback**.

4. Discussion of Findings

- **P1 (Theory):** Small sample but strong performance on most algorithmic techniques. Some weaker areas indicate need for additional support in graph-based algorithms.
- **P2 (Systems):** PhD students met advanced expectations with strong performance across distributed systems assignments.
- **P3 (Languages):** Results reveal gaps in formal methods and logic preparation. Updating prerequisites and course content is recommended.
- **P4 (Research):** Most students passed oral qualifiers, but presentation skills remain an area for improvement.

5. Improvement Plans

Planned Modification	Responsible Person	Implementation Date	Intended Outcome
Encourage students to take CS430 before CS535	Graduate Committee & CS535 Instructor	AY25–26	Strengthen mathematical foundations for theory
Integrate modern cloud/distributed frameworks into CS550	CS550 Instructor	AY25–26	Ensure alignment with current systems research
Update CS401 or add CS330/CS440/CS430 prerequisites for CS536	Graduate Committee & CS536 Instructor	AY25–26	Improve preparation in discrete mathematics and logic
Require practice presentations before qualifiers	Graduate Advisors & Committee	AY25–26	Improve student ability to present research findings clearly

6. Assessment Process Recommendations

- Rubrics continue to be useful for aligning evaluation across PhD core courses.
- Consider early diagnostic assignments (like CS535 “Homework 0”) in other qualifier courses.
- Explore integrating **project-based assessments** to supplement exams and qualifiers.

7. Proposed Assessment Plan for Next Year

Learning Goal	Expected Sample Size	Semester(s) Collected	Rubric Used	Artifact Source	Evaluation Date	Evaluator(s)
P1: Theory	TBD	AY25–26	Theory Rubric	CS535	TBD	TBD
P2: Systems	TBD	AY25–26	Systems Rubric	CS550	TBD	TBD
P3: Languages	TBD	AY25–26	Languages Rubric	CS536	TBD	TBD
P4: Research	TBD	AY25–26	Research/Qualifier Rubric	Oral Qualifier Exam	TBD	TBD

8. Appendices

Appendix A: Rubric for Objective P1

- **Course:** CS535 – Design and Analysis of Algorithms
- **Instructor:** Peng-Jun Wan
- **Evaluator:** Sanjiv Kapoor
- **Semester:** Spring 2025
- **Number of PhD Students:** 3

Criteria / Topic	Description	Weight	Measurement Method	Performance Target	% Students Achieving
Constrained graph traversal	Graph construction, replication, inference via reachability	10%	Written assignment ≥ 85 , quiz/midterm/final ≥ 80	$\geq 80\%$	67%
Shortest paths, widest paths, min-mean circuits	Greedy, DP, relaxation, reweighting, complexity boundary	10%	Written assignment ≥ 85 , quiz/midterm/final ≥ 80	$\geq 80\%$	100%
Backtracking with stack	Linear time with processing, amortized analysis	10%	Written assignment ≥ 85 , quiz/midterm/final ≥ 80	$\geq 80\%$	100%
Speed-up in dynamic programming	Monge, submodularity, monotonicity, charging argument	10%	Written assignment ≥ 85 , quiz/midterm/final ≥ 80	$\geq 80\%$	100%
Decrease and conquer with median	Parametric search, contraction of instances	10%	Written assignment ≥ 85 , quiz/midterm/final ≥ 80	$\geq 80\%$	100%
Minimum spanning tree and arborescence	Hybrid algorithms, contraction/expansion	10%	Quiz/midterm/final ≥ 80	$\geq 80\%$	67%
Bipartite matching and covering	Potential method, augmenting in batch	10%	Written assignment ≥ 85 , quiz/midterm/final ≥ 80	$\geq 80\%$	67%
Max-flow, min-cut, applications	Duality, polynomial complexity classes	15%	Written assignment ≥ 85 , quiz/midterm/final ≥ 80	$\geq 80\%$	67%
Approximation algorithms	Restriction, scaling, primal-dual	15%	Written assignment ≥ 85 , quiz/midterm/final ≥ 80	$\geq 80\%$	100%

Overall Performance: 85% of PhD students achieved the standard. **Discussion:** 2 students received **A**, 1 received **B**. Students developed skills in formulating optimization problems, designing algorithms, and analyzing complexity.

Appendix B: Rubric for Objective P2

- **Course:** CS550 – Advanced Operating Systems
- **Instructor:** Xian-He Sun
- **Evaluator:** Oliver Michel
- **Semester:** Spring 2025
- **Number of PhD Students:** 8

Criteria / Topic	Description	Weight	Measurement Method	Performance Target	% Students Achieving
Distributed systems concepts	Understanding of distributed principles	9%	Five written assignments ≥80	≥80%	92%
Centralized file sharing system	Design & implementation of centralized file sharing	14%	Programming Assignment 1 ≥80	≥80%	95%
Distributed file sharing system	Design & implementation of distributed file sharing	14%	Programming Assignment 2 ≥80	≥80%	92%
Consistency protocols	Implementation of consistency mechanisms	14%	Programming Assignment 3 ≥80	≥80%	92%
Overall knowledge	Comprehensive systems knowledge	28%	Final Exam ≥60 (A required for PhD)	≥80%	87%
Research project	Topic-specific research and report	15%	Final project report ≥75	≥80%	90%

Overall Performance: All PhD students achieved the learning goal, meeting the A-level standard.

Appendix C: Rubric for Objective P3

- **Course:** CS536 – Science of Programming
- **Instructor:** Xiaolang Wang
- **Evaluator:** Yutong Wang
- **Semester:** Spring 2025
- **Number of PhD Students:** 6

Criteria / Topic	Description	Weight	Measurement Method	Performance Target	% Students Achieving
Predicate logic correctness	Verify correctness of logic statements	12%	Homework & Exams	$\geq 70\%$	83.3%
Formal semantics	Determine semantics of sequential programs	6%	Homework & Exams	$\geq 70\%$	66.7%
Hoare triples	Verify validity of correctness triples	10%	Homework & Exams	$\geq 70\%$	66.7%
Proofs of correctness	Verify/manipulate formal proofs	12%	Homework & Exams	$\geq 70\%$	83.3%
Loop correctness	Determine correctness of loops (invariants)	12%	Homework & Exams	$\geq 70\%$	66.7%
Complex assignments	Correctness of complex assignments	8%	Homework & Exams	$\geq 70\%$	66.7%
Substitutions	Perform textual substitutions for proofs	8%	Homework & Exams	$\geq 70\%$	33.3%
Runtime errors	Avoidance of runtime errors	8%	Homework & Exams	$\geq 70\%$	83.3%
Program convergence	Determine program convergence	8%	Homework & Exams	$\geq 70\%$	33.3%
Parallel programs	Correctness of parallel programs	16%	Exams	$\geq 70\%$	50%

Overall Performance: Significant variation, with low achievement in substitutions, program convergence, and parallel programming. **Discussion:** Students lacked adequate discrete mathematics preparation. Recommendations include revising prerequisites (CS330/CS430) or expanding CS401.

Appendix D: Rubric for Objective P4

- **Exam:** Oral Qualifier
- **Evaluator:** Bogdan Korel
- **Semesters:** Fall 2024, Spring 2025
- **Number of PhD Students:** 7 (3 in Fall, 4 in Spring)

Criteria / Topic	Description	Weight	Measurement Method	Performance Target	% Students Achieving
Literature review	Ability to conduct a structured literature review	25%	Oral qualifier exam, presentation, report	≥80%	85%
Research question formulation	Ability to define research problems	25%	Oral qualifier exam, presentation, report	≥80%	85%
Empirical/theoretical support	Ability to collect empirical or theoretical evidence	25%	Oral qualifier exam, presentation, report	≥80%	85%
Presentation of findings	Ability to present research clearly	25%	Oral qualifier exam, presentation, report	≥80%	85%

Overall Performance: 6 of 7 passed (85% success rate). **Discussion:** Weakest areas were research support and presentation of findings. Recommendation: require **practice presentations** with advisor/peer feedback.

Appendix E: Summary of Cumulative Changes from Prior Assessments (PhD)

This appendix summarizes changes made to the PhD in Computer Science program as a result of ongoing assessment cycles. The aim is to show continuous improvement in response to identified student challenges and evolving disciplinary needs.

1. Curriculum Adjustments

- **Expanded qualifier preparation:** Core graduate courses (CS535, CS550, CS536) were more tightly aligned with PhD qualifier expectations, with additional assignments and exam components for PhD students.
- **Research emphasis:** Increased integration of research-oriented projects in advanced courses, particularly in CS550 (systems) and CS536 (languages).
- **Course prerequisites reviewed:**
 - Recommendation to integrate more discrete mathematics/logic content into CS401 or require CS330/CS440/CS430 to better prepare students for CS536.
 - CS430 strongly encouraged before CS535 for PhD theory-track students.

2. Course-Level Changes

- **CS535 (Theory):** Introduced diagnostic assignments (Homework 0) and tailored PhD-level homework/exam questions to strengthen algorithmic rigor and analysis.
- **CS550 (Systems):** Expanded programming assignments from centralized to distributed systems with consistency enforcement; planning to incorporate cloud/distributed frameworks.
- **CS536 (Languages):** Adjusted grading rubrics to better capture mastery of formal semantics, Hoare logic, and program verification; proposed prerequisite revisions to address gaps in discrete mathematics.

3. Assessment and Evaluation Process

- **Rubrics developed for all PhD core courses** (CS535, CS550, CS536) and for the Oral Qualifier (P1–P4), providing greater consistency and transparency.
- **Oral qualifier evaluation** made more structured, with explicit categories (literature review, problem formulation, support, presentation).
- **Standardized higher thresholds for PhD students** (e.g., requiring A-level performance in CS550) compared to MS students.

4. Program Standards and Admissions

- **Raising admissions standards:** Discussions on strengthening baseline math and logic preparation for incoming PhD students, particularly those targeting the theory and languages tracks.
- **Preparation monitoring:** Increased use of early diagnostic tools to identify students who may need remedial coursework before sitting for qualifiers.

5. Research Training Enhancements

- **Practice presentations introduced:** Students now encouraged to give trial talks before oral qualifiers, with advisor and peer feedback.
- **Greater emphasis on communication skills** in research milestones, addressing recurring weaknesses in presenting results.
- **Encouraging interdisciplinary projects:** Students are increasingly steered toward projects that cross systems, theory, and applications to better prepare for cutting-edge research.

6. Strategic Initiatives

- Development of a **multi-year roadmap for qualifier reform**, ensuring that learning goals (P1–P4) remain well integrated with course outcomes and evolving research frontiers.
- **Faculty coordination** through the Graduate Committee ensures that changes to prerequisites, rubrics, and assessment processes are consistently reviewed and implemented.

